BodyWise

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what problem does it solve?

-Many people are unaware of whether their body weight is within a healthy range and often lack access to simple tools that can help them understand their health status. As a result, they may overlook potential health risks associated with being underweight, overweight, or obese. Additionally, individuals often struggle to find basic nutrition and health recommendations that are relevant to their body condition.

-This project addresses this problem by providing a BMI-based health assessment system that calculates a user's Body Mass Index (BMI), determines their body condition category, and offers personalized nutrition and health tips. The system helps users quickly evaluate their physical health status, promotes awareness of healthy lifestyle habits, and supports informed decisions regarding diet, exercise, and overall well-being.

Why did this project was chosen?

-This project was chosen because maintaining a healthy body weight is an important part of overall health, yet many people do not know how to assess their body condition or understand the health implications of their weight. A BMI-based health assessment system provides a simple and effective way to evaluate an individual's weight status and raise awareness about healthy living.

-Additionally, the project combines several practical concepts such as user input processing, calculations, decision-making, and personalized recommendations, making it both educational and useful. By providing nutrition and health tips based on BMI results, the system helps users make more informed lifestyle choices and encourages healthier habits.

-Overall, this project was selected because it addresses a real-world health concern while demonstrating the application of programming to solve everyday problems.

```python
def initUi(self):
    # Widgets (texts, buttons..)
    self.secondT = QLabel(secondT)
    self.getWeight = QLineEdit()
    self.h = QLabel(hei)
    self.getHeight = QLineEdit()
    self.continu = QLabel(click_check_result
    self.botn = QPushButton(b2n)
    #Layout + adding the widgs
    self.v_line = QVBoxLayout()
    self.v_line.addWidget(self.secondT)
    self.v_line.addWidget(self.getWeight)
    self.v_line.addWidget(self.h)
    self.v_line.addWidget(self.getHeight)
    self.v_line.addWidget(self.continu)
    self.v_line.addWidget(self.botn)
    # layout and 2nd windo
    self.setLayout(self.v_line)
```

As shown in thi block of code here the user enters his weight and height + setting a layout for the window and applying the widgets to be shown.

```python
    # User data
    self.weight = float(weight)
    self.height = float(height)
    # BMI calculation
    self.bmi = self.weight / (self.height ** 2)
```

Here we do the calculation for the bmi based on the data that the user inputs.

```python
# BMI condition checking
if self.bmi < 18.5:
    self.condition = "Underweight"
elif self.bmi < 25:
    self.condition = "Normal weight"
elif self.bmi < 30:
    self.condition = "Overweight"
else:
    self.condition = "Obesity"
```

Here we check the body condition based on the bmi.

What was the hardest part?

-The difficulty was handling user input correctly, such as ensuring that the user enters valid height and weight values before performing the BMI calculation. It was also necessary to implement conditional logic to classify BMI results into different body condition categories and display the appropriate nutrition and health recommendations.

What did i learn?

-Through this project, I learned how to develop a desktop application using PyQt5 and integrate a graphical user interface with Python code. I gained experience in creating and organizing GUI components such as buttons, labels, text fields, and layouts.

What would you improve or add with more time ?

-If I had more time, I would improve the user interface to make it more attractive and easier to use. I would also add features such as personalized meal plans, exercise recommendations, progress tracking, and graphs to visualize BMI changes over time. Additionally, I would include more health metrics and better error handling to provide a more complete and reliable health assessment experience.

Thanks

Thank you for your attention.
Before concluding, I would like to thank our teacher for all the effort, patience, and support provided during this course. The knowledge and guidance you shared played an important role in helping me complete this project and improve my programming skills.
Thank you for your time, and I hope you enjoyed my presentation.